

QUIZ 2

Answer all questions. Submit your answer in elearning on Wednesday 14 January 2026 not later than 5 pm.

1.

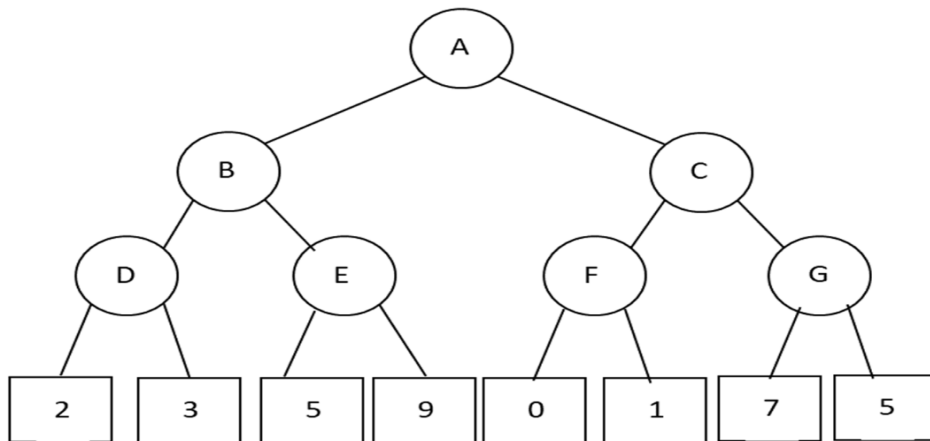


Figure 1

- Based on the given state space search in Figure 1, which of the node will be pruned?
- Using the AlphaBeta Pruning Algorithm, What is the correct **Alpha, α** and **Beta, β** value for **node C**?
- What is the heuristic value of **node B**?

2.

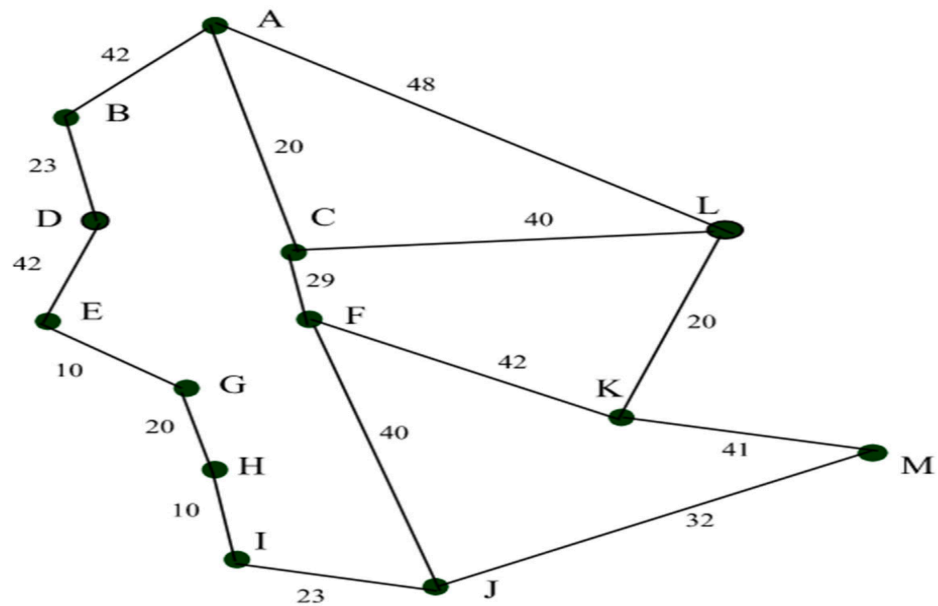


Figure 2

Using the A* algorithm, work out a route from town A to town M as shown in Figure 2. Use the following cost functions.

$G(n)$ = The cost of each move as the distance between each town (shown on map).

$H(n)$ = The Straight-Line Distance between any town and town M. These distances are given in the table below.

Straight Line Distance to M

A	51	E	42	I	50	M	0
B	50	F	14	J	32		
C	32	G	33	K	41		
D	28	H	43	L	56		

Provide the search tree for your solution, showing the order in which the nodes were expanded and cost at each node. You should not re-visit states previously visited. Finally, state the route you would take and the cost of that route.

3. Consider the game three shown in Figure 3 below and answer the following questions.

Perform a left to right alpha beta pruning on the tree. Indicate all parts of the tree that are pruned. Provide an explanation to support your decision. Redraw this game tree with the solution.

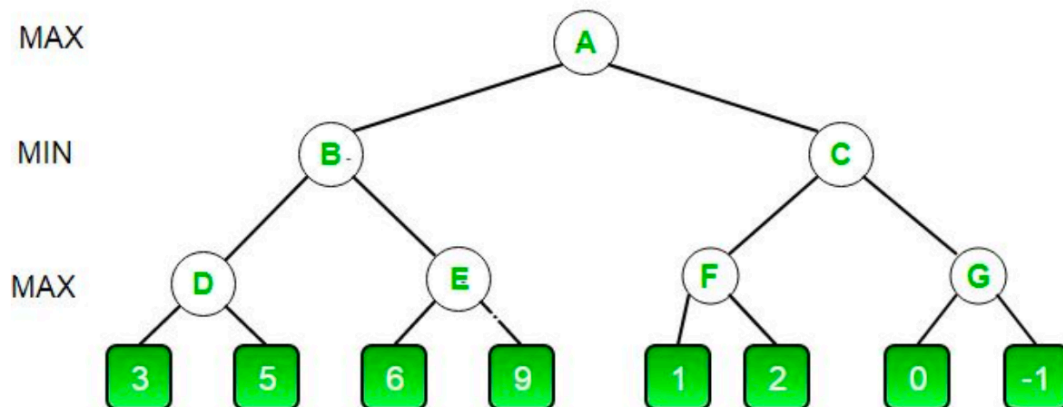


Figure 3

4. The following rules are knowledge base of a production system. Use this rule set to answer questions i).

- Rule 1 : IF Fred is trained
THEN Fred is a good cat.
- Rule 2: IF a cat is a good cat AND he has a master
THEN he will be with his master
- Rule 3: IF it is a warm day
THEN his master is at the park
- Rule 4: IF it is cold day
THEN his master is at the café
- Rule 5 : IF Fred is with his master AND in his master is at the café
THEN Fred is at the cafe

Use a **DATA DRIVEN SEARCH** to describe the production system table including its working memory, conflict set and rule fired.

The initial data given is :

Fred is trained

Fred has master

It is a cold day

And the goal is :

Fred is at the cafe

Terminate when the goal i.e **Fred is at the café** is in the working memory. Refer

Table 1 for symbol. Show your answer as shown in Table 2.

Note : Use the symbol as in Table 1 to fill-up your answer in Table 2.

Table 1

A	:	Fred is trained
B	:	Good cat
C	:	Has a master
D	:	Will be with his master
E	:	Warm day
F	:	His master at the park
G	:	Cold day
H	:	His master at the café
I	:	Fred is at the café
J	:	Fred is at the park

Table 2

Iteration #	Working memory	Conflict set	Rule fired
0			